Keziah K. Adjei

PhD Candidate | Chemical and Biological Engineering | Iowa State University
kka55@iastate.edu | LinkedIn | Orcid | 267-265-6320

Professional Summary

Interdisciplinary PhD Candidate in Chemical and Biological Engineering with expertise spanning immunology, molecular medicine, diagnostics, biomaterials, and bioengineering. Experienced in developing innovative immune profiling platforms, diagnostic assays, and translational research technologies that bridge engineering and biology. Proven ability to lead independent research projects, collaborate across multidisciplinary teams, mentor junior researchers, and communicate scientific findings to diverse technical and non-technical audiences. Passionate about advancing scientific innovation through translational research, consumer health, therapeutic development, diagnostics, and biomedical technologies that improve patient and consumer outcomes.

Education

PhD in Chemical and Biological Engineering | Iowa State University, Ames
Expected Completion – Spring 2028
Advisor: Molly Kozminsky, Assistant Professor
Research Focus: Immunoengineering | Spatially resolved cytokine capture technologies | Biomaterials and hydrogel engineering | Single-cell immune profiling |Assay development for translational immunology

MSc in Molecular Medicine | Drexel University, Philadelphia
Academic Excellence in Molecular Medicine, 2024
Advisor: Mary Ann Comunale, Associate Professor
Research Focus: Glycosylation biomarkers | Translational diagnostics | Lyme disease biomarker discovery | Immunological assay development | Glycoproteomics

BSc in Biotechnology and Molecular Biology | University for Development Studies, Tamale, Ghana
Best Graduating Female Student, 2019
Advisor: Francis Addy, Senior Lecturer
Research Focus: Molecular diagnostics | Parasitology |Veterinary/public health | Molecular epidemiology

Technical, Clinical and Research Skills

Immunology & Cell Biology
Mammalian cell culture | Immune cell differentiation | Dendritic & T cell biology | Immune profiling | Immunofluorescence & IHC | Cell labeling technologies

Molecular Biology & Diagnostics
ELISA | Multiplex lectin assays | RT-PCR | RNA extraction | Antibody conjugation | Biomarker analysis | Fluorescence microscopy | Diagnostic assay development

Bioengineering & Platform Development
DNA-directed patterning | Spatial biology | Biomaterials engineering | Assay development | Biosensor development | Platform validation | Microfabrication

Instrumentation
Fluorescence microscopy | SpectraMax iD3 | Luminex 200 | Image Analysis tools – ImageJ, GraphPad Prism, Licor Image Studio | Autocad |

Scientific and Professional Skills
Translational Research | Cross-functional collaboration | Problem solving | Critical thinking | Leadership | Mentorship | Technical training | Process improvement | Organization | Scientific communication | Experimental troubleshooting

Research Experience

Graduate Research Assistant | Chemical and Biological Engineering, Iowa State University, Ames, IA | May 2024 – Present

Key Contributions:

1. Lead the development of a DNA-directed patterned platform to investigate functional heterogeneity in adaptive immune cell interactions through spatially controlled dendritic cell–T cell pairing.
2. Design, fabricate, and optimize biomaterial-based single-cell platforms integrating DNA-directed patterning, hydrogel engineering, and immune cell culture to investigate localized immune responses.

3. Develop cytokine capture assays for spatially resolved detection of immune mediators to improve understanding of immune cell communication.
4. Engineer antibody-oligonucleotide conjugation strategies and optimize capture chemistries to enhance assay sensitivity, specificity, and reproducibility.
5. Perform mammalian immune cell culture, differentiation, activation, and characterization using immunological and fluorescence-based techniques.
6. Design and execute independent experiments, analyze multidimensional datasets, troubleshoot technical challenges, and implement workflow improvements to increase experimental reliability.
7. Develop laboratory protocols, SOPs, and documentation supporting reproducible experimental workflows and high-quality research practices.
8. Mentor undergraduate researchers through technical instruction, laboratory supervision, experimental planning, and scientific communication.
9. Present research findings during laboratory meetings, departmental seminars, national conferences, and collaborative research discussions..

Research Assistant | College of Medicine, Drexel University, Philadelphia, PA | August 2022 – May 2024
Key Contributions:
1. Investigated glycosylation biomarkers associated with Lyme disease through multidisciplinary translational research integrating molecular biology, glycobiology, and immunology.
2. Designed, optimized, and validated multiplex lectin ELISA assays for characterization of immunoglobulin glycosylation patterns associated with disease progression.
3. Collaborated with interdisciplinary research teams to design experiments, interpret complex datasets, and translate findings into potential diagnostic applications.
4. Authored laboratory SOPs, maintained research documentation, and contributed to manuscript preparation and scientific presentations.
5. Trained new laboratory personnel in laboratory techniques, instrumentation, safety practices, and experimental procedures.
6. Applied critical thinking and systematic troubleshooting to optimize assay performance and improve experimental reproducibility.

Research Assistant | Department of Virology, Noguchi Memorial Institute, Accra, Ghana | 2019 – 2021
Key Contributions:
1. Performed high-throughput molecular diagnostic testing for SARS-CoV-2 using RT-PCR and viral RNA extraction techniques while maintaining rigorous laboratory quality standards.
2. Supported Ghana's national COVID-19 response through timely and accurate molecular diagnostic testing.
3. Evaluated commercial molecular diagnostic kits and RT-PCR platforms to support validation studies and regulatory assessment activities.
4. Participated in WHO Inter-Laboratory Proficiency Testing programs to ensure laboratory accuracy, quality assurance, and compliance with international standards.
5. Developed laboratory SOPs and contributed to standardized molecular diagnostic workflows.
6. Assisted in antimicrobial resistance educational initiatives supporting public health awareness and laboratory capacity building.
7. Trained approximately twenty laboratory personnel in molecular diagnostic techniques, biosafety procedures, laboratory quality practices, and instrument operation.
8. Collaborated with multidisciplinary scientists, clinicians, and public health professionals supporting national disease surveillance initiative

Undergraduate Researcher | Molecular Biology, University for Development Studies, Ghana | 2018 – 2019
Key Contributions:
1. Performed genomic DNA extraction, PCR amplification, and PCR-RFLP analysis to differentiate Fasciola species.
2. Analyzed molecular data to support epidemiological characterization of parasitic infections in livestock.
3. Strengthened laboratory skills in molecular diagnostics, experimental design, data interpretation, and scientific reporting.

Professional Experience
Teaching Assistant
Iowa State University
1. Facilitate undergraduate laboratory instruction and provide technical guidance in chemical and biological engineering concepts.
2. Support student learning through laboratory demonstrations, assessment, and scientific communication.

Laboratory Manager

Drexel University and Noguchi Memorial Institute for Medical Research

1. Coordinated laboratory operations, including equipment management, inventory control, procurement, workflow organization, and laboratory maintenance.
2. Ensured laboratory readiness while supporting efficient research and diagnostic operations.

Professional Affiliations

- Nanovaccine Institute, Iowa State University – 2024 – Present
- Rising Doctoral Institute, Iowa State University – 2024 – Present
- American Institute of Chemical Engineers (AIChE) member, 2025 – Present
- American Society of Tropical Medicine and Hygiene (ASTMH), 2022 – Present
- Biotechnology Students Association of Ghana, 2015 – 2019
- International Association of Agriculture & Related Sciences (2015 – 2019)

Research Interests

Consumer Health Innovation | Immunology | Drug Discovery | Therapeutic Development | Translational Medicine | Biomarker Discovery | Diagnostics | Cell Engineering | Single-Cell Technologies | Bioengineering | Clinical Development | Assay Development

---

Selected Publications

Keziah Kwarteng Adjei, Andrea Boyso Salinas, and Molly Kozminsky. DNA Patterned Secretome Capture Array for Spatially Resolved Single Cell Cytokine Mapping (In peer review – Lab on a chip journal)

Adjei, K.K., Salinas, A.B., Kozminsky, M. In Vitro Patterned Platform to Study T-Cell Activation Through Spatially Defined Cell–Cell Interactions**.** Manuscript in Preparation.

Addy, F., Narh, J.K., **Adjei, K.K**., Adu-Bonsu, G. *Dicrocoelium* spp. in cattle from Wa, Ghana: prevalence and phylogeny based on 28S rRNA. *Parasitol Res* 120, 1499–1504 (2021). https://doi.org/10.1007/s00436-021-07085-z

Addy, F., Boafo, K.F., Yakubu, A.B., Turkson, K.A, Narh, J.K., **Adjei, K.K**., Adu-Bonsu, G. *Taenia hydatigena* in goat and sheep in Ghana: a cross-sectional abattoir survey in Northern and Upper West Regions. *J Parasit Dis* (2021). https://doi.org/10.1007/s12639-020-01331-4

Leadership and Service

Vice President
Graduate Student Association – Drexel University

- Represented graduate students across interdisciplinary programs while fostering professional development and community engagement.
- Organized networking and career development initiatives that strengthened student engagement and collaboration.

Laboratory Manager – Noguchi Memorial Institute for Medical Research

- Coordinated laboratory operations, inventory management, equipment maintenance, and workflow organization to support efficient research activities.

Undergraduate Research Mentor – Iowa State University

- Mentor undergraduate researchers in laboratory techniques, scientific communication, experimental planning, and research best practices.

Awards and Honors

- SMDP Biotech Scholar - 2026
- Peter J. Reilly Graduate Scholarship – 2024
- Frederick Martison Chemical Engineering Fund – 2024
- Academic Excellence in Molecular Medicine – Drexel University - 2024
- Fulbright Foreign Student Scholarship – 2022-2024
- PEO International Peace Scholarship award – 2022-2024
- Van Duuren Travel Grant Award – 2023
- Hungaricum Stipendium scholarship – 2021
- Ghana Bilateral Scholarship for Masters – 2021
- Noguchi Memorial Institute – Virology Department – Outstanding performance plaque
- Best graduating Female Student, University for Development Studies – 2019
